Reference contributor / ACE Engineering — GoM Subsea Well Intervention & Maintenance

Client-ready brief · generated 2026-06-27

worldenergydata ▸ Capabilities ▸ Insights ▸ Subsea interventions

Subsea well intervention — Gulf of Mexico supply-constraint brief

Takeaway

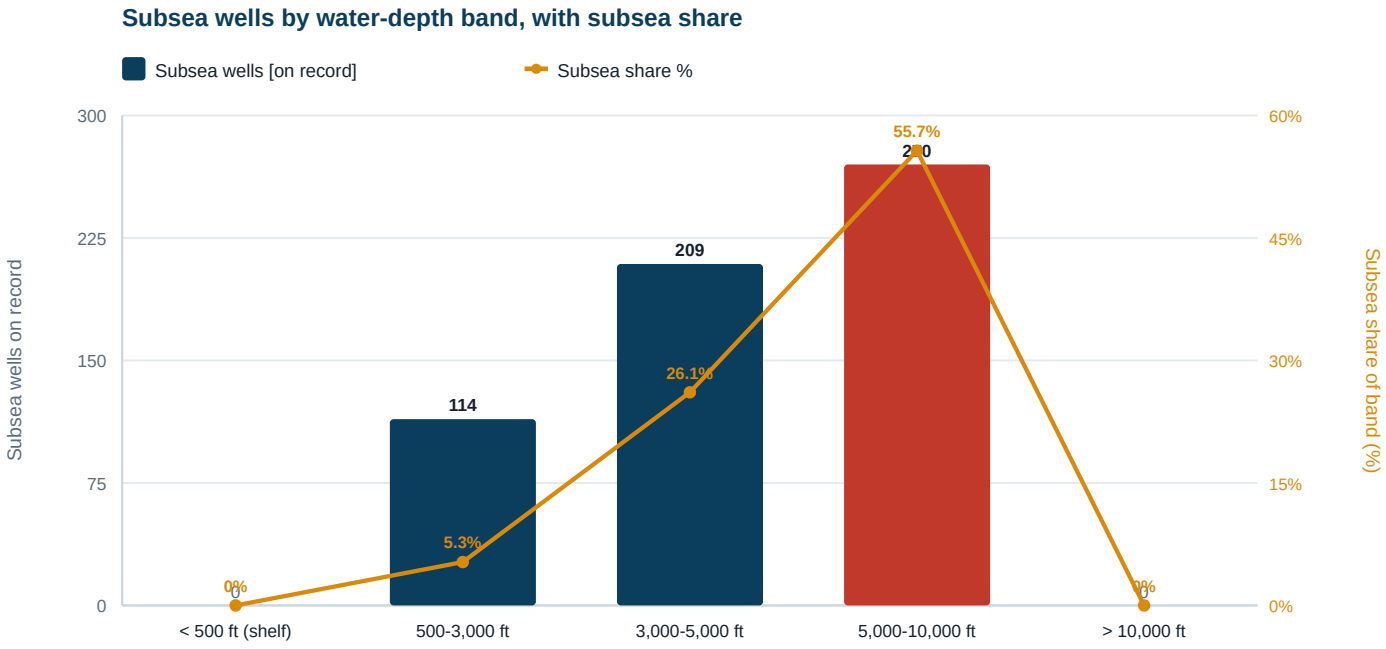
Supply-constrained. The deepest serviceable band (5,000-10,000 ft) holds the most subsea wells on record (270 wells, 55.7% subsea share) yet sits over the thinnest dedicated fleet: only ~3 GoM-resident dedicated intervention units [soft]. The wells are concentrated exactly where the intervention-asset supply is scarcest.

Existing subsea wells by water-depth band

Counts are from the BSEE subsea-borehole registry [on record]; the subsea-share column depends on full-population coverage and is [soft].

Band	Subsea wells [on record]	Wells in band	Subsea share [soft]
< 500 ft (shelf)	0	25,844	0.0%
500-3,000 ft	114	2,156	5.3%
3,000-5,000 ft	209	801	26.1%
5,000-10,000 ft	270	485	55.7%
> 10,000 ft	0	0	n/a

Subsea total on record: **593** wells [on record] (depth min 1,055 ft / median 4,609 ft / max 9,627 ft; n=593). Well population compared against: 29,286.



Subsea wells and subsea-share invert with depth — the deepest serviceable band holds the most subsea wells (270) at 55.7% share.
source: well_inventory_by_band.yml (#583)

Serviceability — who can reach each well

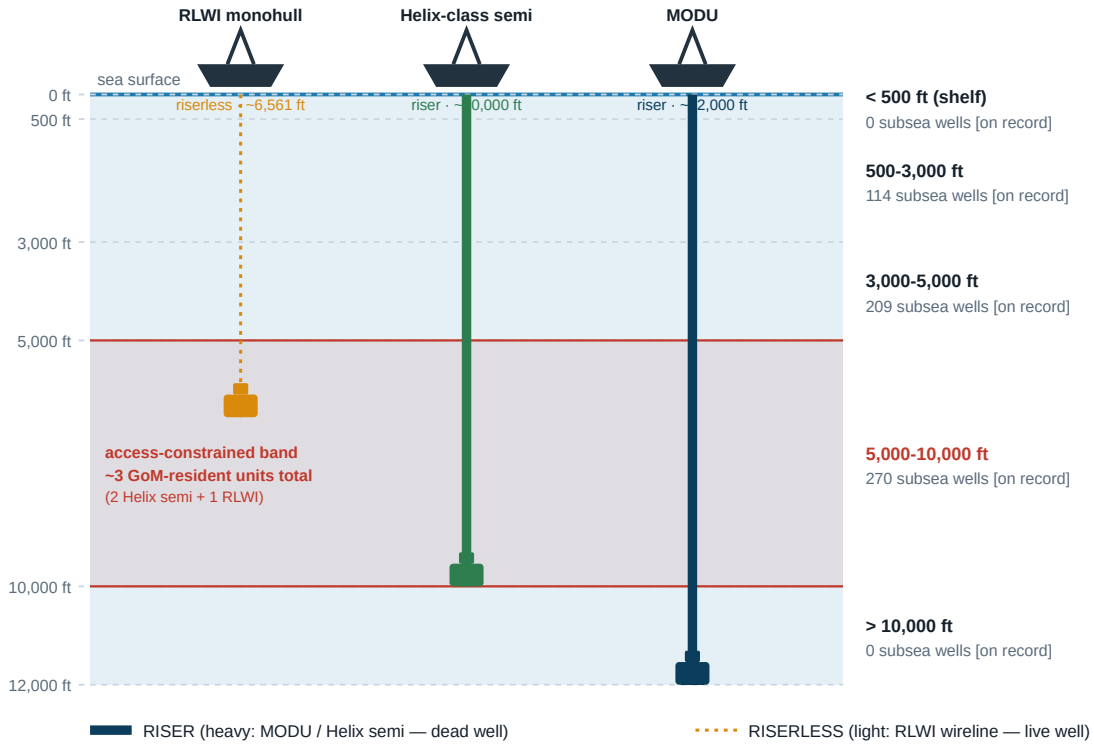
The rule: Depth does not pick the asset; riser-need picks the asset. Water depth alone does not choose the vessel; whether the scope needs an intervention/workover riser does. These are [engineering judgement] from standard subsea-intervention practice, not a measured BSEE dataset.

Band	Scope	Riser required	Eligible asset classes
< 500 ft (shelf)	Light live-well (riserless wireline)	no	rlwi_monohull, mpsv
< 500 ft (shelf)	Through-tubing coiled tubing	yes	heavy_intervention_semi, modu
< 500 ft (shelf)	Heavy dead-well (tubing pull / recompletion / P&A)	yes	modu, heavy_intervention_semi
500-3,000 ft	Light live-well (riserless wireline)	no	rlwi_monohull, mpsv
500-3,000 ft	Through-tubing coiled tubing	yes	heavy_intervention_semi, modu
500-3,000 ft	Heavy dead-well (tubing pull / recompletion / P&A)	yes	modu, heavy_intervention_semi
3,000-5,000 ft	Light live-well (riserless wireline)	no	rlwi_monohull, mpsv
3,000-5,000 ft	Through-tubing coiled tubing	yes	heavy_intervention_semi, modu
3,000-5,000 ft	Heavy dead-well (tubing pull / recompletion / P&A)	yes	modu, heavy_intervention_semi
5,000-10,000 ft	Light live-well (riserless wireline)	no	rlwi_monohull, mpsv (availability-constrained)
5,000-10,000 ft	Through-tubing coiled tubing	yes	heavy_intervention_semi, modu
5,000-10,000 ft	Heavy dead-well (tubing pull / recompletion / P&A)	yes	modu, heavy_intervention_semi
> 10,000 ft	Light live-well (riserless wireline)	no	rlwi_monohull, mpsv
> 10,000 ft	Through-tubing coiled tubing	yes	heavy_intervention_semi, modu
> 10,000 ft	Heavy dead-well (tubing pull / recompletion / P&A)	yes	modu, heavy_intervention_semi

Modifiers: A horizontal subsea tree (HXT) tubing pull forces a riser-based asset even for otherwise-light scope. Light riserless work in 5,000-10,000 ft is availability_constrained (thin supply of rated units), not capability-limited.

Asset classes: `rlwi_monohull` = Riserless light well intervention monohull (vessel_fleet #592); `mpsv` = Multipurpose support vessel (vessel_fleet #592); `heavy_intervention_semi` = Heavy-intervention semi, e.g. Helix Q-class (vessel_fleet #592); `modu` = Mobile offshore drilling unit (drillship / drilling semi).

Depth-band serviceability: who can reach each well



Cross-section of who can reach each depth: riserless RLWI monohull (live well) vs riser-based Helix semi / MODU (dead well); only ~3 dedicated units are GoM-resident.
source: well_inventory_by_band.yml (#583), serviceability_matrix.yml (#586), intervention_fleet_catalog.yml (#598)

Intervention-asset fleet

GoM-resident dedicated intervention units (the binding supply)

~3 units [soft] are GoM-resident and dedicated to well intervention. This — not any global roster — is the figure that constrains GoM well access:

Unit	Class	Intervention	Water-depth rating (m)
Helix Q4000	heavy_intervention_semi	heavy	3,048
Helix Q5000	heavy_intervention_semi	heavy	3,048
Island Performer	rlwi_monohull	light	2,000

Triangulated against independent research:

- gom resident dedicated intervention vessels: **2-4** [soft] — GoM-resident dedicated intervention units; small population
- global rlwi fleet: **15-25** [soft] — global riserless light well-intervention fleet, +/-30-50%
- helix purpose built units: **7-8** [soft] — Helix purpose-built well-intervention units worldwide
- rigs rated 20k psi worldwide: **2** [on record] — only two rigs worldwide rated to 20,000 psi

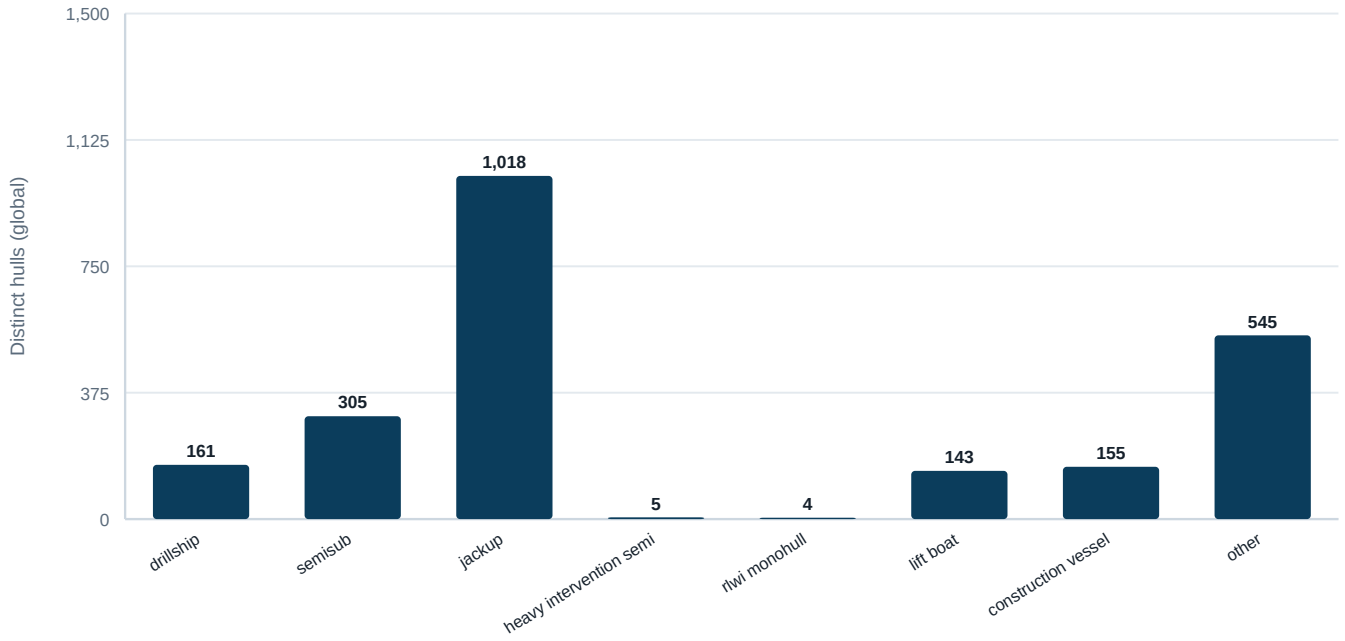
Global available roster (CONTEXT ONLY — NOT GoM supply)

These are WORLDWIDE fleet counts. They include shelf and non-GoM assets and must NOT be read as Gulf-of-Mexico intervention supply. They show the size of the global pool a GoM operator competes for, nothing more. Counts are DEDUPED distinct hulls (#599): 2,442 source listings collapse to 2,336 vessels (106 duplicates removed, e.g. the Helix Q-class name-spelling variants), so no hull is double-counted.

Indicative day-rates are an attached PUBLIC snapshot (#596), as of 2026-02-19 — issuer FSRs / filings / dated trade press, NOT a live subscription feed. Helix Q-class intervention semis and RLWI monohulls carry no public per-day figure (shown as dayrate not public).

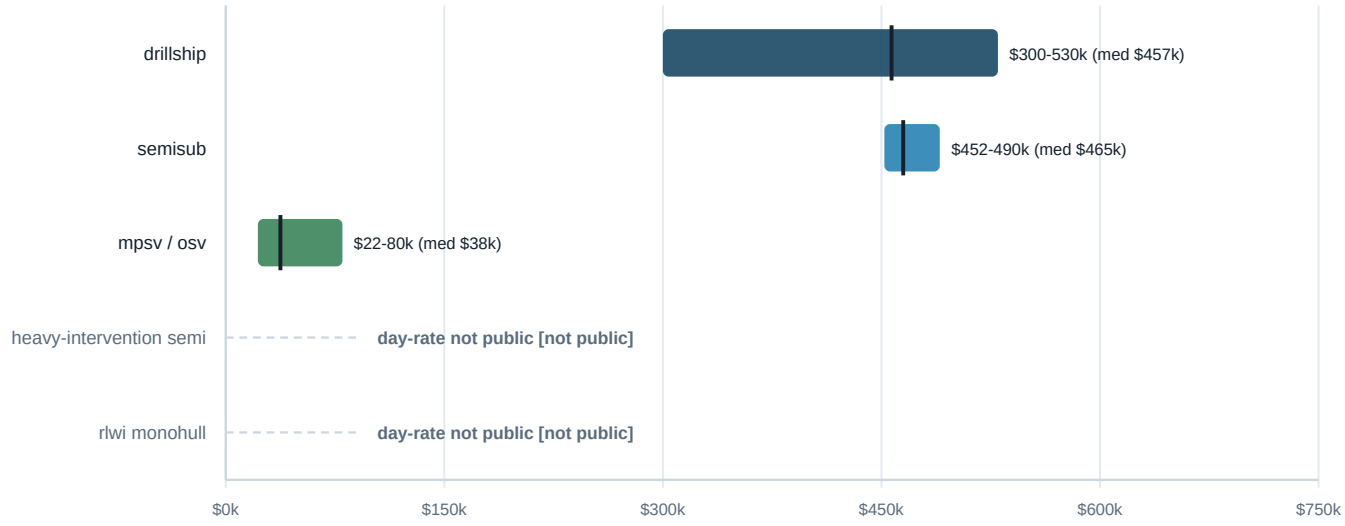
Asset class (global)	Count [deduped]	Water-depth capability (ft)	Indicative day-rate
modu_drillship	161	4,900-12,000	$300,000-$530,000/day (median $457,000) [analyst, on_record, reported]
modu_semisub	305	1,500-12,500	$452,000-$490,000/day (median $465,000) [on_record]
modu_jackup	1,018	300-492	n/a
heavy_intervention_semi	5	1,197-10,000	dayrate not public [not public]
rlwi_monohull	4	6,561-6,561	dayrate not public [not public]
lift_boat	143	n/a	n/a
construction_vessel	155	49-13,123	n/a
other	545	5,000-5,000	n/a

Global available roster by asset class (deduped distinct hulls)



Deduped global roster by class (CONTEXT only, not GoM supply): 1,018 jackups vs only 5 heavy-intervention semis and 4 RLWI monohulls. **source: intervention_fleet_catalog.yml (#598/#599)**

Indicative day-rate bands by asset class ($k/day)



Indicative public day-rate bands; intervention semis and RLWI monohulls carry no public per-day figure.
source: intervention_fleet_catalog.yml (#596), intervention_economics.yml (#629)

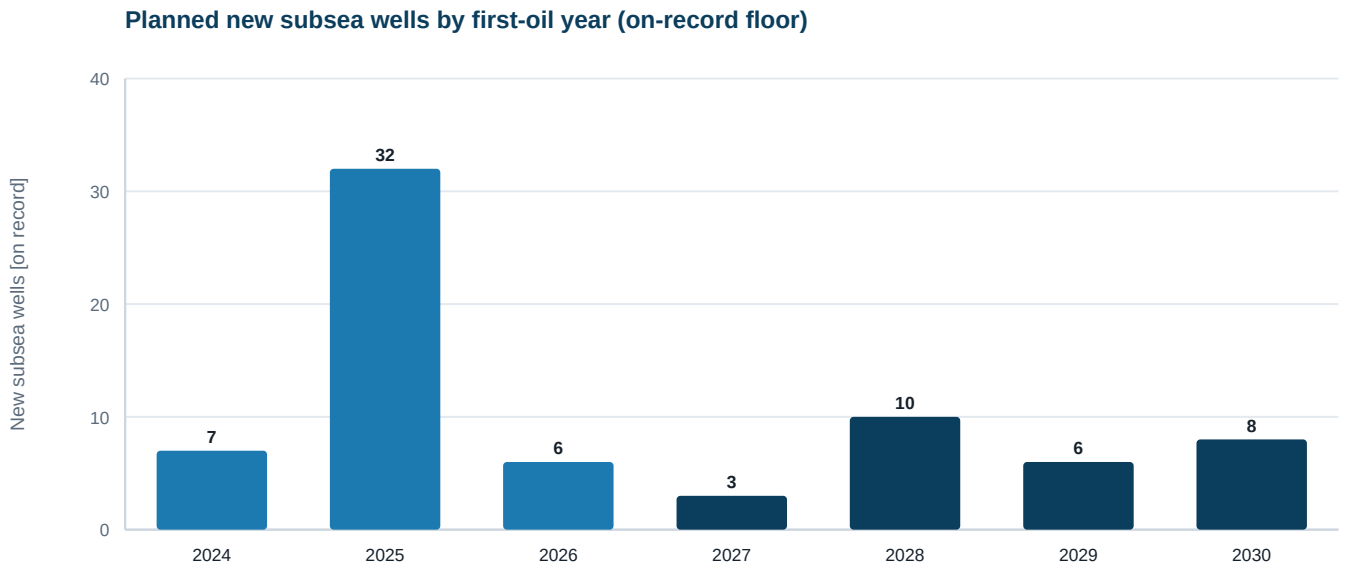
Planned / projected new subsea wells

The on-record FID register itemises **18 announced US GoM developments** totalling **72 new subsea wells [on record]** — a FLOOR, since later infill phases are deliberately not itemised. Trion is excluded (Mexican waters).

New wells by first-oil year (on-record floor):

First-oil year	New subsea wells [on record]
2024	7
2025	32
2026	6
2027	3
2028	10
2029	6
2030	8

Analyst envelope: **12-20 new GoM subsea trees / wells installed per year** (central 16, +/-15%) [projected], through 2030. This brackets the on-record floor (P10/P90); it is not summed with it. Sources: Westwood Subsea Tree Tracker (GoM ~16-19 trees/yr); Rystad Energy subsea market outlook.



On-record FID floor of new subsea wells by first-oil year (7/32/6/3/10/6/8) — later infill phases are deliberately excluded.
source: planned_subsea_wells.yml (#587)

Access gap — forward-looking demand vs supply

FORWARD-LOOKING ACCESS RISK, not a measured crossover. Per band: intervention demand rig-days/yr (subsea wells x parameterized frequency x band-effective heavy-intervention duration) vs available fleet rig-days/yr (eligible fleet x utilization x 365). A utilization_ratio > 1 means forward demand would outstrip that supply pool; it is an access-risk indicator, not an observed shortfall.

5,000-10,000 ft holds 270 subsea wells but only 2 GoM-resident heavy-intervention unit(s): forward demand would need ~4.4x the rig-days that resident supply can provide (access-RISK indicator, NOT a measured crossover).

Per band: intervention DEMAND rig-days/yr (subsea wells x intervention frequency ~0.1-0.2/well/yr [engineering judgement] x band-effective heavy-intervention duration, #629) vs available SUPPLY rig-days/yr (eligible fleet x utilization ~0.6-0.8 [engineering judgement] x 365). The GoM-RESIDENT fleet is the binding constraint; the global roster is CONTEXT ONLY. utilization ratio = demand / supply (>1x = forward demand outstrips that supply pool). All figures are [forward-looking risk] — an access-RISK indicator, NOT a measured crossover.

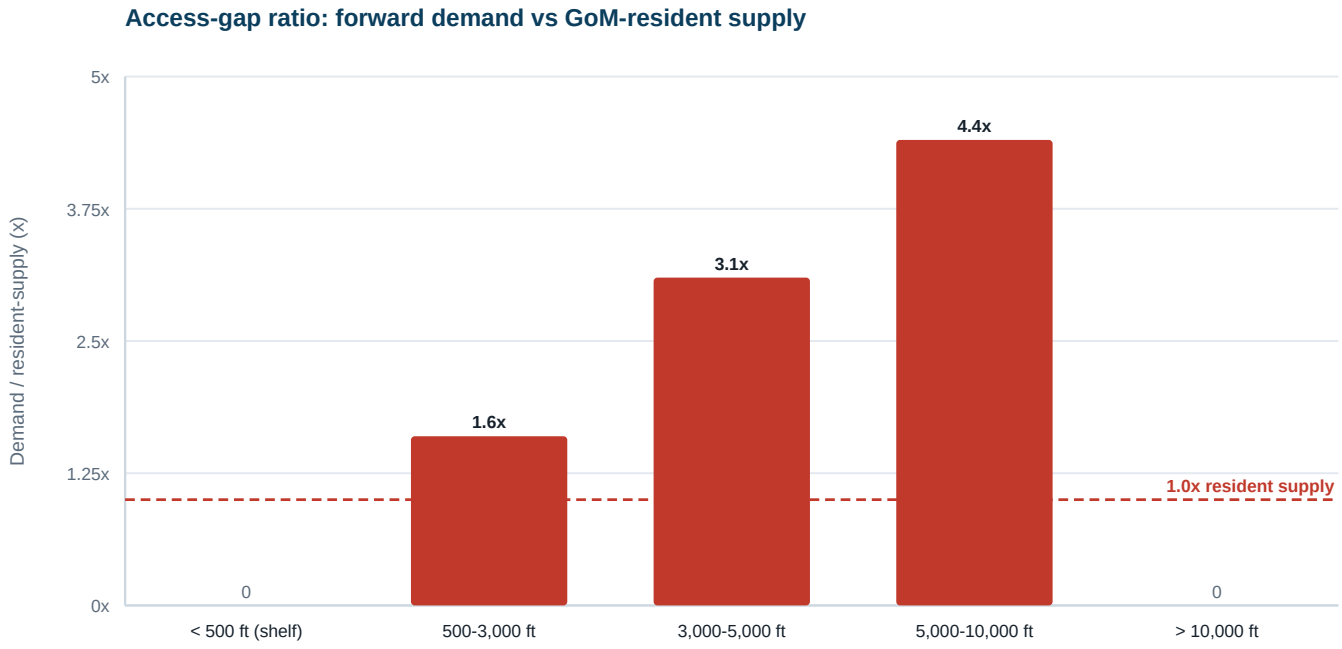
Band	Subsea wells [on record]	Demand rig-days/yr [forward]	GoM-resident heavy units	GoM-resident supply rig-days/yr	Demand/supply ratio (GoM-resident) [forward]	$ exposure/yr [forward]
< 500 ft (shelf)	0	0	2	511	0.0x	n/a
500-3,000 ft	114	821	2	511	1.6x	$375.1M
3,000-5,000 ft	209	1,599	2	511	3.1x	$730.7M
5,000-10,000 ft	270	2,228	2	511	4.4x	$1.0B
> 10,000 ft	0	0	0	0	n/a	$0.0M

Global-roster context (NOT GoM supply): against the worldwide heavy-deepwater pool (~471 eligible units) the 5,000-10,000 ft demand ratio is only ~0.0x — the global pool is ample; the constraint is that almost none of it is GoM-RESIDENT. This is the access wall.

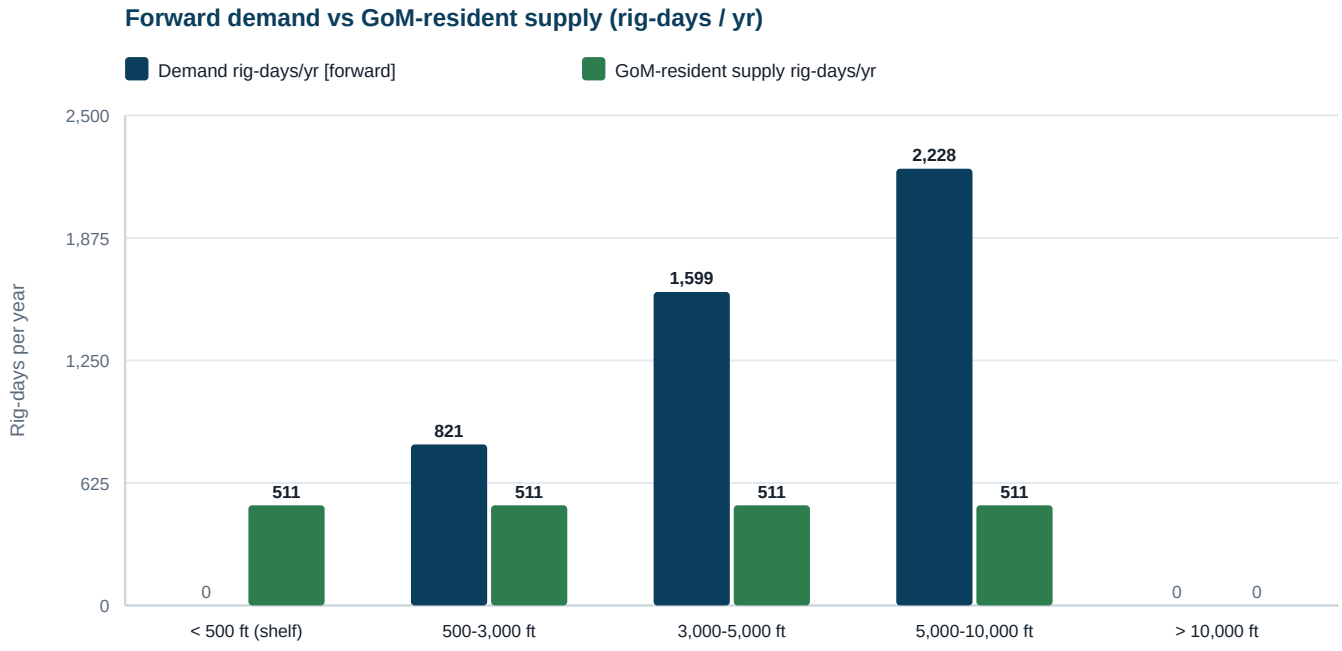
Forward-looking caveats carried with the synthesis:

- FORWARD-LOOKING: this is access RISK, not a measured demand/supply crossover; both parameters (frequency, utilization) are cited planning assumptions, overridable.
- FLOOR: the #630 empirical interventions_per_well cross-check is a lifetime upper proxy on the ~6% depth-stamped WAR subset (WATER_DEPTH populated on only ~6% of WAR rows, #627/#630), not an annual lambda.
- GEOGRAPHY (#628): GoM has ~0 RESIDENT dedicated LIGHT-intervention vessels -- this is geography (the global RLWI fleet is elsewhere) + the 6% WAR coverage, NOT evidence that deepwater light intervention is avoided. ~3 GoM-resident dedicated units total (#598/#628).

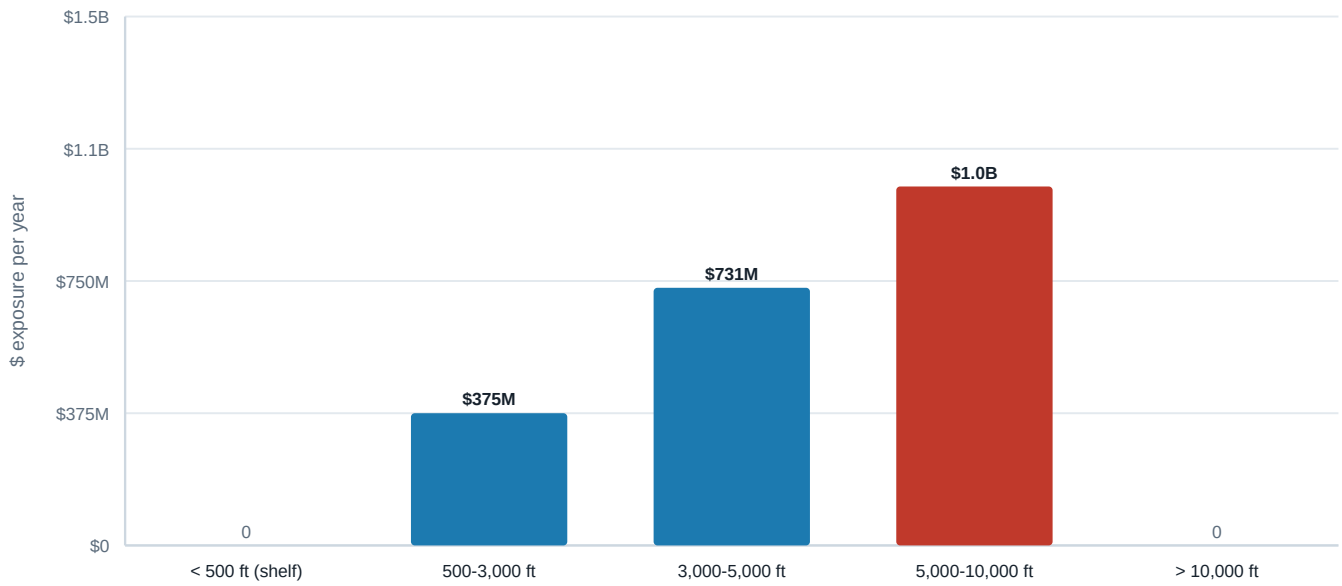
- UNKNOWN 44.5% (#627): a large share of WAR service-type is unclassified, so the empirical work mix is incomplete.
- Demand uses the HEAVY heavy_dead_well duration for all interventions (an upper bound on rig-days); real campaigns mix lighter scopes.
- Global-roster supply is a worldwide pool a GoM operator competes for; the GoM-resident view is the binding constraint.
- Subsea well counts are [on record] and likely undercount older completions (#583/#589).



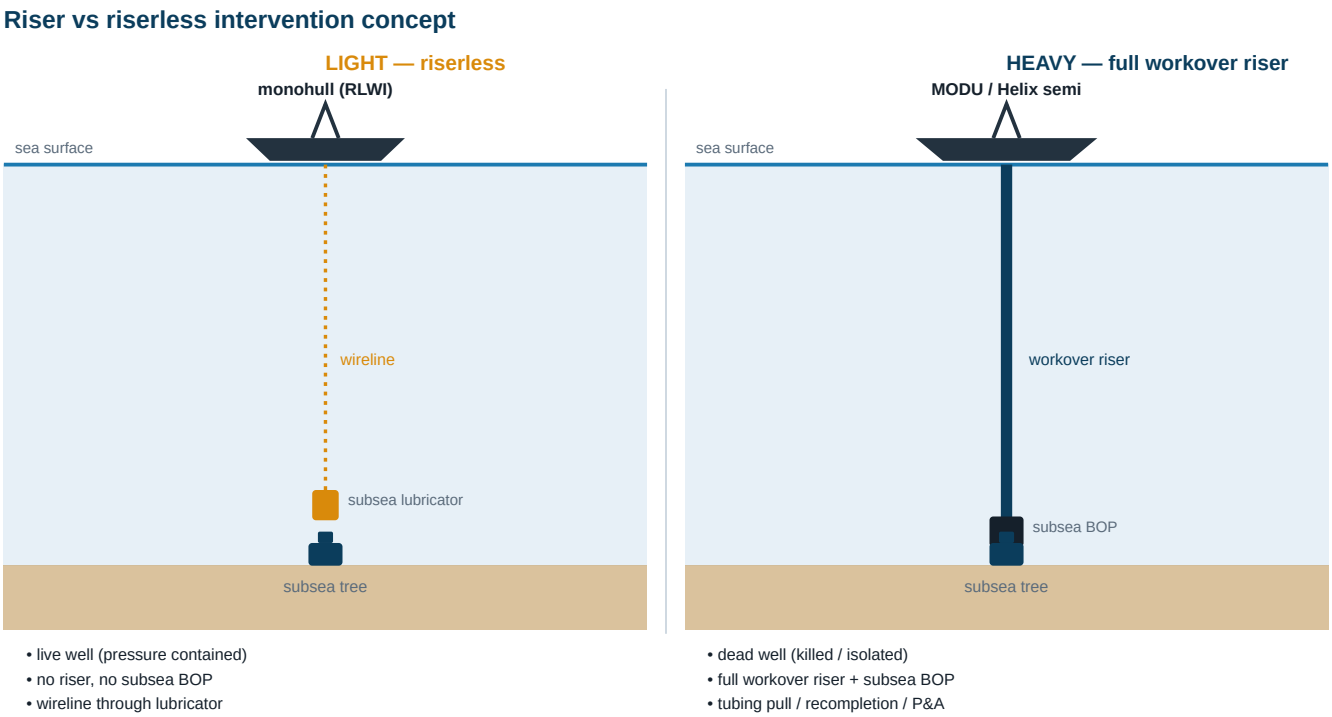
Forward demand vs GoM-resident supply ratio: 1.6x / 3.1x / 4.4x in the populated deepwater bands (>1.0x = access risk, NOT a measured crossover). **source: access_gap.yml (#638)**



Central forward demand rig-days/yr vs the fixed ~511 rig-days/yr the GoM-resident heavy fleet can provide.
source: access_gap.yml (#638)



Central annual $ exposure by band: $375M / $731M / $1.0B (forward-looking planning indicator). **source: access_gap.yml (#638)**



The capability split behind the gap: light riserless (monohull, live well) vs heavy riser-based (MODU/Helix, dead well).
source: serviceability_matrix.yml (#586)

Honest-framing notes

- Subsea band counts are [on record] from the BSEE borehole registry, which likely undercounts older completions; subsea-share is [soft].
- Serviceability is [engineering judgement], not a measured dataset; band-level availability should be calibrated against the fleet.
- The global by-asset-class fleet counts are CONTEXT only. The binding GoM intervention supply is ~3 resident dedicated units (~2-4 [soft]).
- The planned-wells total is an on-record FLOOR; the analyst rate is [projected] and brackets — never adds to — that floor.
- The access gap is FORWARD-LOOKING access RISK, not a measured crossover; intervention frequency and fleet utilization are cited, overridable planning parameters, and intervention cost now varies by depth band (#629 mobilization fix).

Sources

All figures render from these committed source YAMLs:

- `well_inventory_by_band.yml` — installed subsea wells by band (worldenergydata #583).
- `serviceability_matrix.yml` — band x scope serviceability + riser rule (worldenergydata #586).
- `planned_subsea_wells.yml` — on-record FID register + analyst rate (worldenergydata #587).
- `intervention_fleet_catalog.yml` — unified intervention fleet (worldenergydata #598).

- `access_gap.yml` — forward-looking demand vs supply rig-days + $ exposure by band (worldenergydata #638, capstone of #626); economics are now band-aware (#629) and demand cross-checks #630.

Generated 2026-06-27
sources: committed YAMLs #583/#586/#587/#596/#598/#629/#630/#638